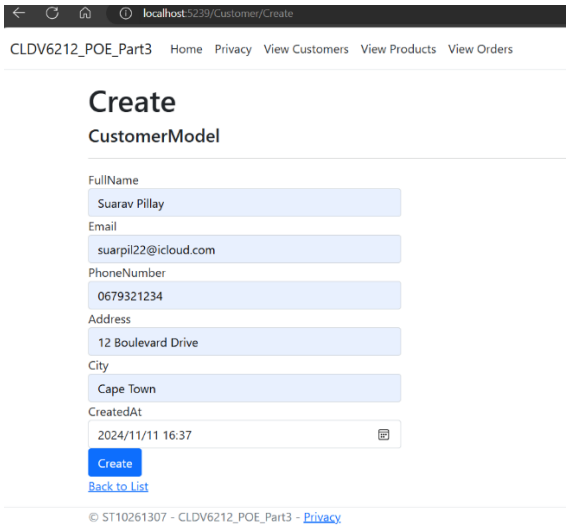
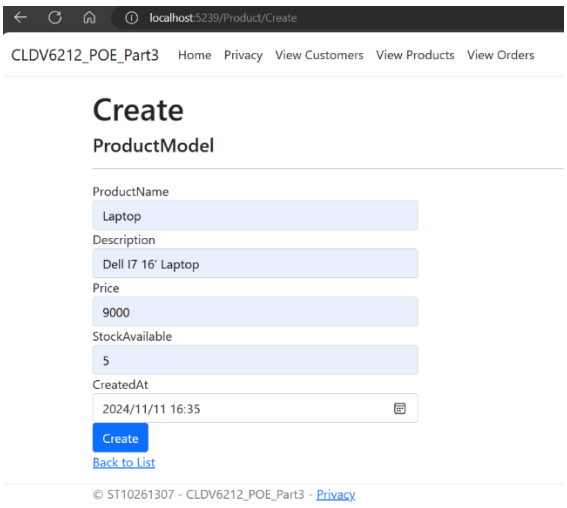
Mohamed Tawfiq Cassim
Retail App – Cloud Development

App Overview:

The application is a cloud-based web application developed using ASP.NET MVC and deployed on Microsoft Azure, designed to manage customer, product, and order information through a scalable cloud architecture. The web application is hosted using Azure App Service, which provides a managed platform for deploying and running the MVC application online, while structured application data such as customer details, product information, and order records are stored in Azure SQL Database, a fully managed relational database service. The system integrates Azure Blob Storage to handle large unstructured data (eg. images/documents) for efficient file storage/retrieval. Azure Functions are used to perform serverless background tasks allowing for automation. Azure Hub and Azure Service Bus are implemented in the architecture for messaging and data streaming between the system components to ensure reliable communication. Database replication is used to improve reliability/performance by duplicating db instances that act as backups, while Azure Backup provides automated protection against data loss. Azure Traffic Manager is used to manage and route incoming user traffic efficiently, sending users to the most appropriate region. Development and testing of the application involve tools such as Visual Studio, GitHub for version control and deployment, and Postman for testing API endpoints and Azure Functions, resulting in a modern cloud-native system that demonstrates the use of multiple Azure Platform-as-a-Service (PaaS) and serverless technologies to create a scalable, reliable, and efficient web-based order management platform.

App Screenshots:





localhost:5239/Customer/Index

CLDV6212_POE_Part3 Home Privacy View Customers View Products View Orders

Index

Create New

FullName	Email	PhoneNumber	Address	City	CreatedAt	
Tawfiq Cassim	mazdarx7@gmail.com	0213456789	2 Waterfall Drive	Johannesburg	2024/11/11 16:23:00	Edit | Details | Delete
Mathew Jacob	matjac@gmail.com	0346789342	5 Lincoln Road	Durban	2024/11/11 16:32:00	Edit | Details | Delete
Suarav Pillay	suarpil22@icloud.com	0679321234	12 Boulevard Drive	Cape Town	2024/11/11 16:37:00	Edit | Details | Delete

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localhost:5239/Product/Index

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Create New

ProductName	Description	Price	StockAvailable	CreatedAt	
Table	2200x1800(mm) Dining Room Table	5000,00	10	2024/11/11 16:29:00	Edit | Details | Delete
Airfryer	Kenwood Dual Cooking Airfryer	4500,00	20	2024/11/11 16:30:00	Edit | Details | Delete
Laptop	Dell I7 16' Laptop	9000,00	5	2024/11/11 16:35:00	Edit | Details | Delete

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Results Messages

Search to filter items...

CustomerID	FullName	Email	PhoneNumber	Address	City	CreatedAt
1	Tawfiq Cassim	mazdarx7@gmail.com	0213456789	2 Waterfall Drive	Johannesburg	2024-11-11T16:23:00.000...
2	Mathew Jacob	matjac@gmail.com	0346789342	5 Lincoln Road	Durban	2024-11-11T16:32:00.000...
3	Suarav Pillay	suarpil22@icloud.com	0679321234	12 Boulevard Drive	Cape Town	2024-11-11T16:37:00.000...

Results Messages

Search to filter items...

ProductID	ProductName	Description	Price	StockAvailable	CreatedAt
1	Table	2200x1800(mm) Dining Room ...	5000.00	10	2024-11-11T16:29:00.0000000
2	Airfryer	Kenwood Dual Cooking Airfryer	4500.00	20	2024-11-11T16:30:00.0000000
3	Laptop	Dell I7 16' Laptop	9000.00	5	2024-11-11T16:35:00.0000000

localhost:5239/Order/Create

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Create

OrderModel

CustomerID

1

OrderDate

2024/11/11 22:09

TotalOrderAmount

14000

Status

Processing

Create

Back to List

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Create New

OrderDate	TotalOrderAmount	Status	Customer
2024/11/11 22:09	14000,00	Processing	Tawfiq Cassim

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Results Messages

Search to filter items...

OrderID	CustomerID	OrderDate	TotalOrderAmount	Status
1	1	2024/11/11 22:09	14000,00	Processing

```
C:\Users\tawfi\AppData\Local\AzureFunctionsTools\Releases\4.89.0-inprocess\cli_x64\func.exe

Azure Functions Core Tools
Core Tools Version:       4.0.6280 Commit hash: N/A +421f0144b42047aa289ce691dc6db4fc8b6143e6 (64-bit)
Function Runtime Version: 4.834.3.22875

[2024-10-01T19:15:06.609Z] Found C:\Users\tawfi\OneDrive\Documents\Visual Studio 2022\Work\CLDV6212_POE_Part2\CLDV6212_P
OE_Part2\CLDV6212_POE_Part2.csproj. Using for user secrets file configuration.

Functions:

        ProcessTheOrders: [POST] http://localhost:7272/api/ProcessTheOrders

        StoreProfile: [POST] http://localhost:7272/api/StoreProfile

        UploadBlobFile: [POST] http://localhost:7272/api/UploadBlobFile

        UploadTheFileShare: [POST] http://localhost:7272/api/UploadTheFileShare

For detailed output, run func with --verbose flag.
[2024-10-01T19:15:13.500Z] Host lock lease acquired by instance ID '000000000000000000000000004CB6AB79'.
```

http://localhost:7272/api/StoreProfile

POST http://localhost:7272/api/StoreProfile

Params Authorization Headers (7) Body Scripts Settings

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers (4) Test Results 200 OK 756 ms 186 B

Pretty Raw Preview Visualize Text

```
1  The users data has been successfully stored.
```

Deployed Website:

https://st10261307.azurewebsites.net/Customer/Index

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Create New

FullName	Email	PhoneNumber	Address
Tawfiq Cassim	mazdarx7@gmail.com	0213456789	2 Waterfall Drive